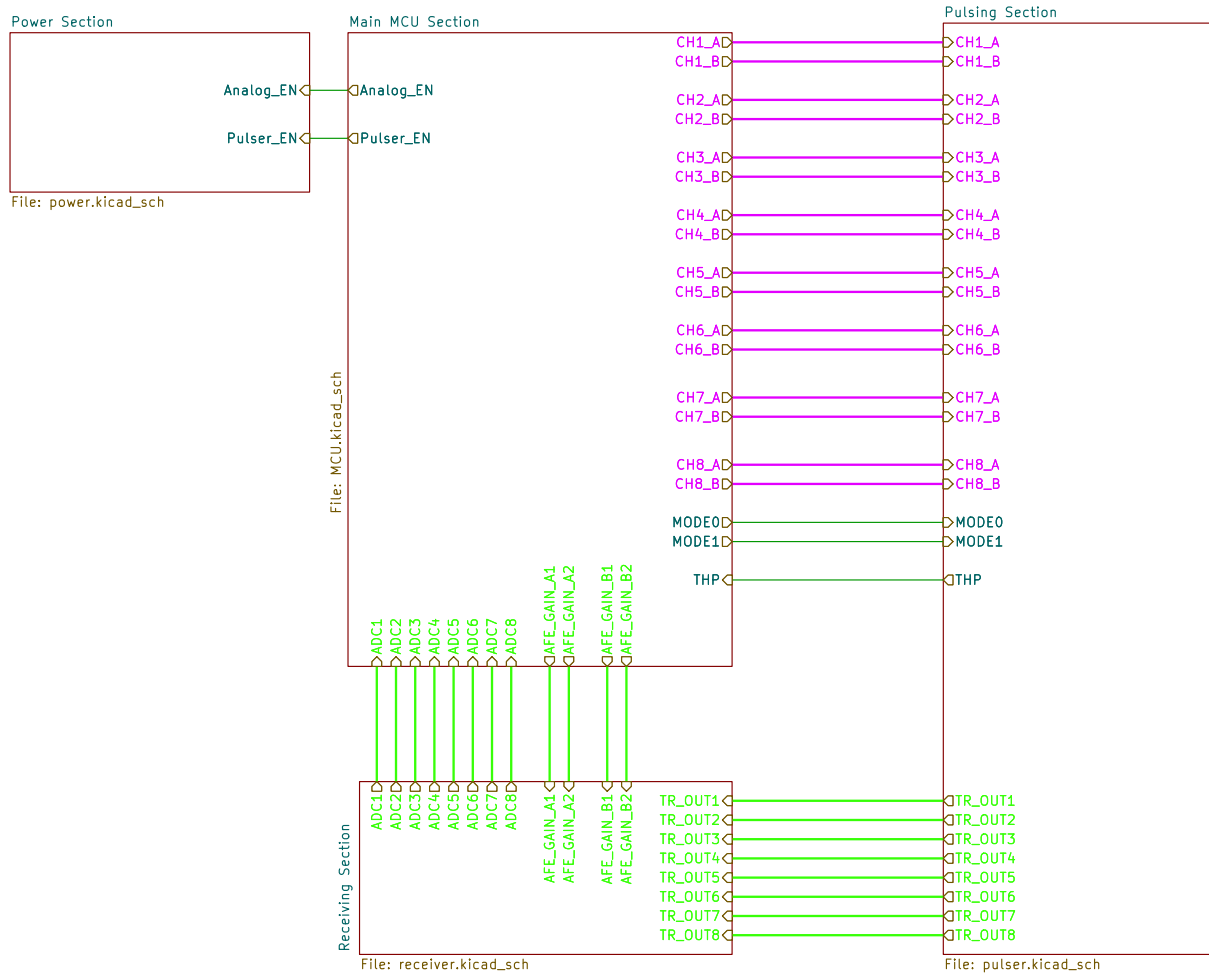
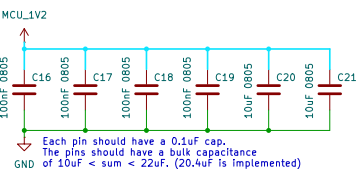
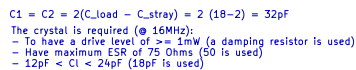
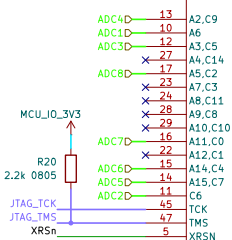
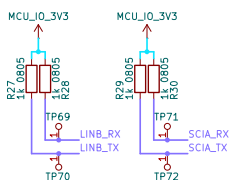
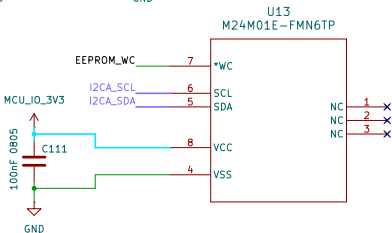




12 Volt (External) (555mA)	7 Volt (Buck Converter) (350mA)	5 Volt (LDO Regulator) (200mA)	LNA + VGA Supply (160mA – short circuit)
			Op-Amps (4 * 10mA = 40mA)
	5 Volt (Buck Converter) (205mA)	3.3 Volt (LDO Regulator) (150mA)	MCU Supply (150mA)
		Positive Pulse Driver Supply (100mA)	
		3.3 Volt (LDO Regulator) (5mA)	Pulser Logic Supply (5mA)
		-5 Volt (Inverting Buck Converter) (100mA)	Negative Pulse Driver Supply (100mA)

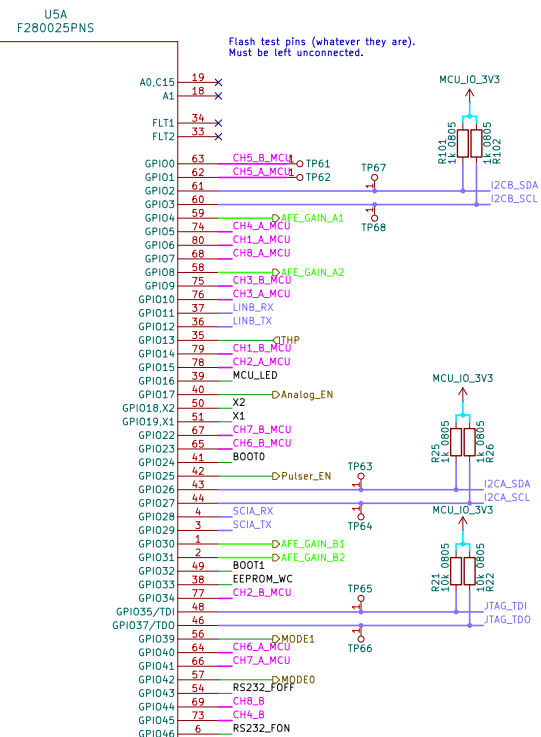
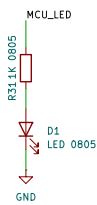


BOOT MODE	GPIO28 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI/UART boot	0	1
CAN	1	0
Flash	1	1

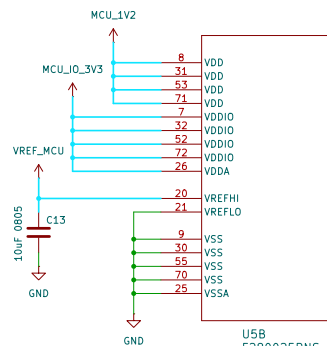


JTAG test-mode select (TMS) with internal pullup. This serial control input is clocked into the TAP controller on the rising edge of TCK. This device does not have a TRSTn pin. An external pullup resistor (recommended 2.2 k Ω) on the TMS pin to VDDIO should be placed on the board to keep JTAG in reset during normal operation.

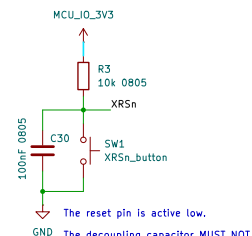
The debugger to be used is the XDS110 debugger (recommended by TI).



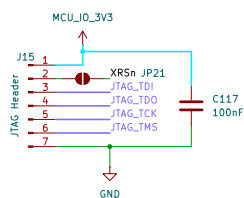
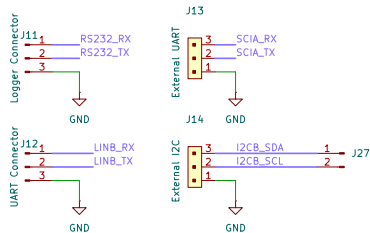
1K resistors are placed at pulser lines to limit current to 3.3mA per pin.



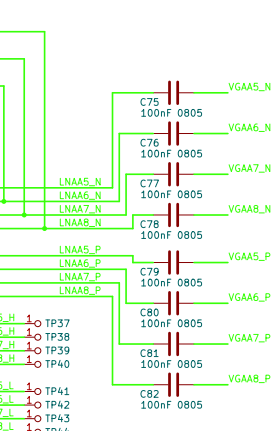
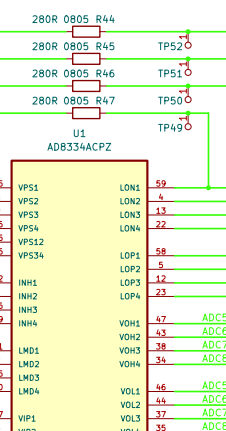
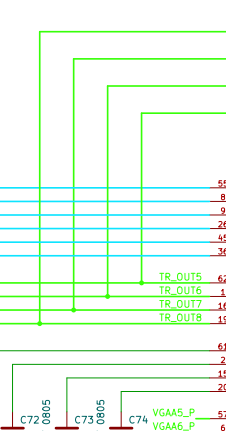
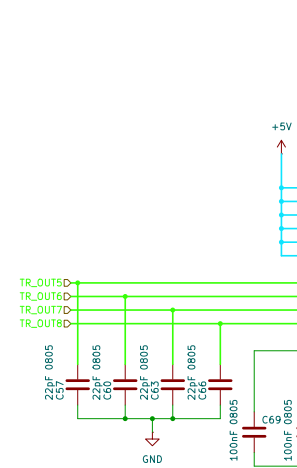
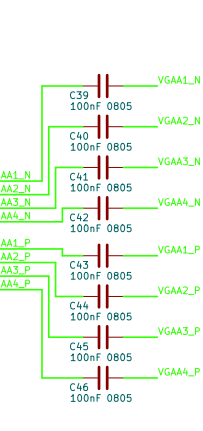
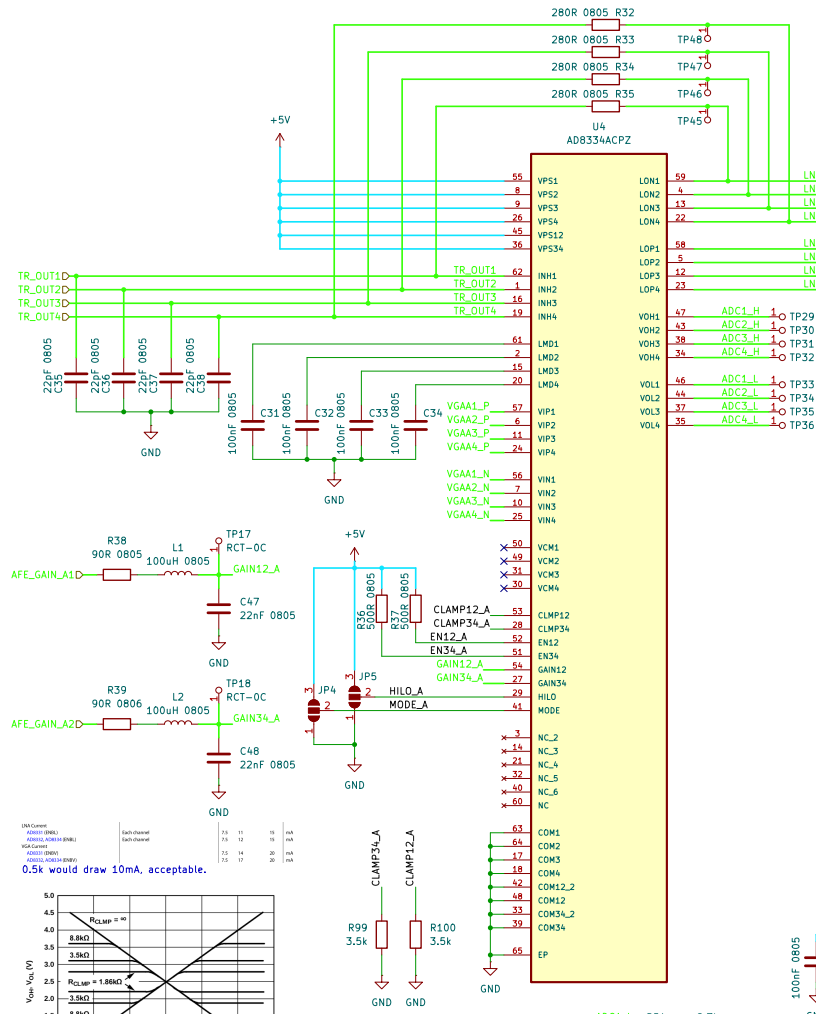
At least 2.2uF is recommended at the Vref_hi pin, 10uF is used.



- The reset pin is active low.
- The decoupling capacitor **MUST NOT** be greater than 100uF or else it will interfere with the watchdog reset functionality.



The MCU has no dedicated JTAG reset pin (TRSTn) but theoretically this could work.



Load Current	Each Channel	Each Channel	Each Channel	Each Channel
400mV (RMS)	7.5	11	15	2.5
400mV (RMS) (100k)	7.5	11	15	2.5
400mV (RMS) (100k)	7.5	11	15	2.5
400mV (RMS) (100k)	7.5	11	15	2.5

0.5k would draw 10mA, acceptable.

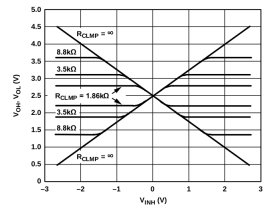
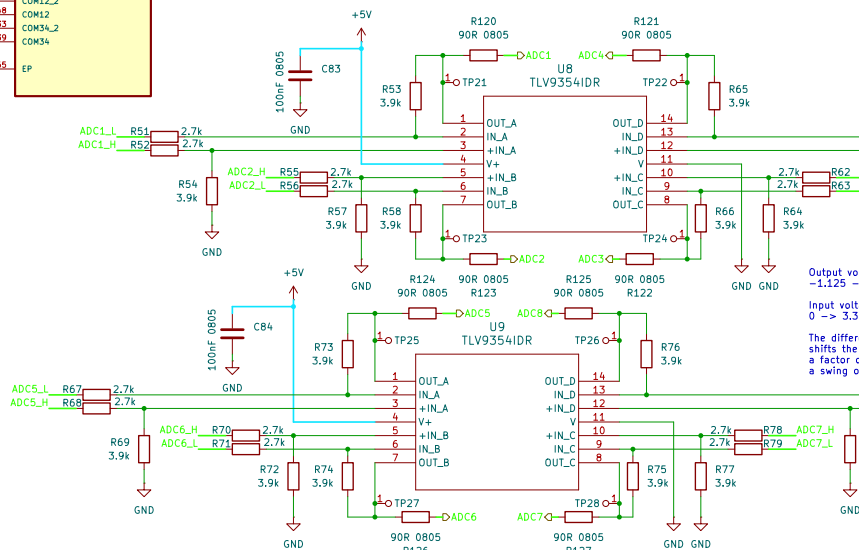


Figure 80. Output Clamping Characteristics

Table 7. LNA External Component Values for Common Source Impedances		
R _{in} (Ω)	R _{in} (Closest STD 1% Value, Ω)	C _{in} (pF)
50	280	22
75	412	12
100	560	8
200	1.13 k	1.2
500	3.01 k	None
6k	>	None

2nd order filter values were recommended by a TI application note (spraa88a), specs:
 - PWM frequency = 5 MHz
 - Cut-off frequency = 107.3 KHz
 - Damping ratio = 0.675 (over-damped but response time = 3 uSec)

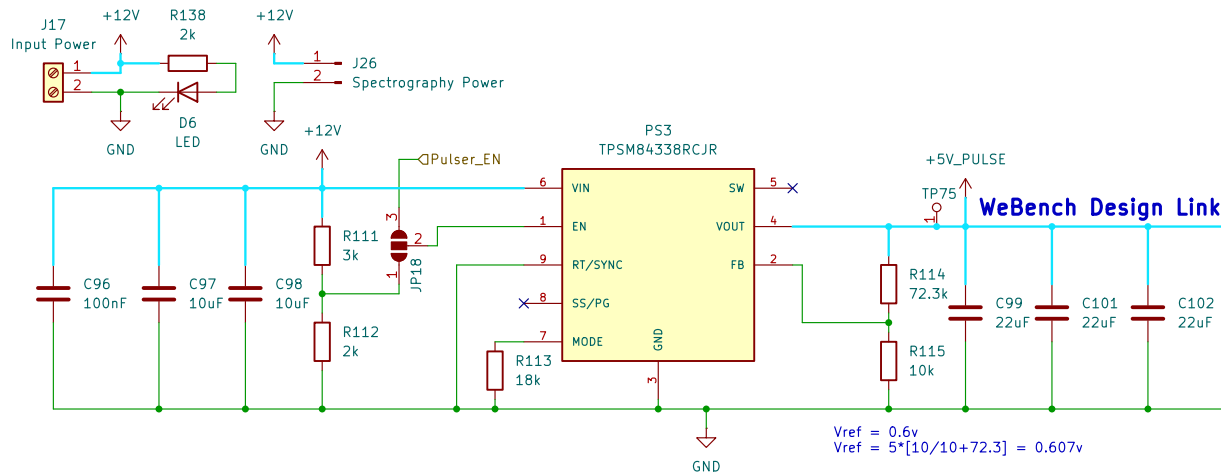
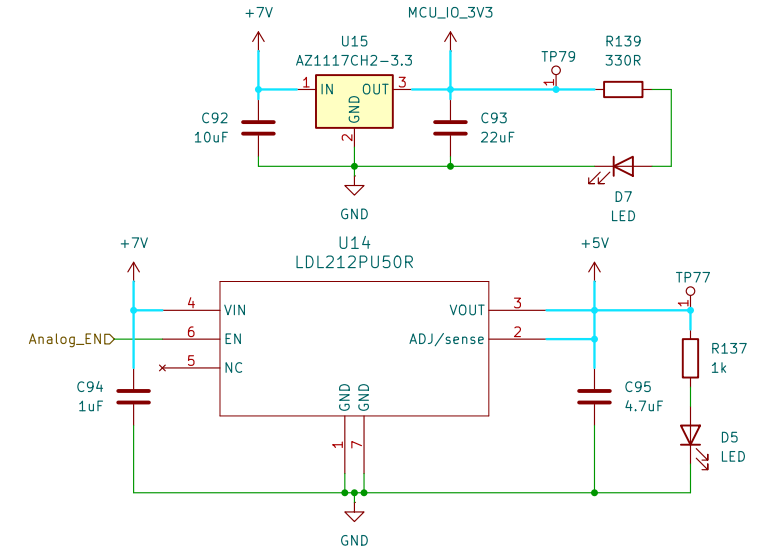
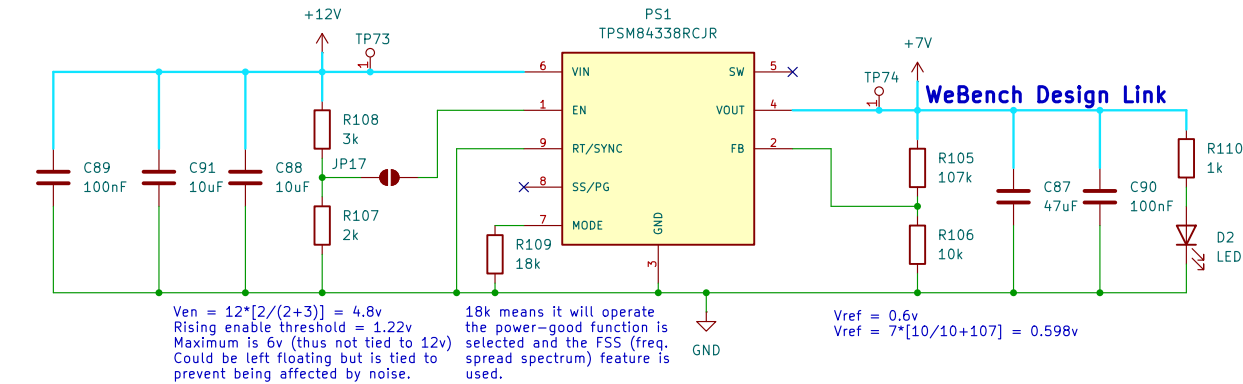
MODE INTERFACE (PIN MODE)		
Logic Level for Positive Gain Slope	0	1.0 V
Logic Level for Negative Gain Slope	2.25	5 V
HELD GAIN RANGE INTERFACE (PIN HELD)		
Logic Level to Select HI Gain Range	2.25	5 V
Logic Level to Select LO Gain Range	0	1.0 V



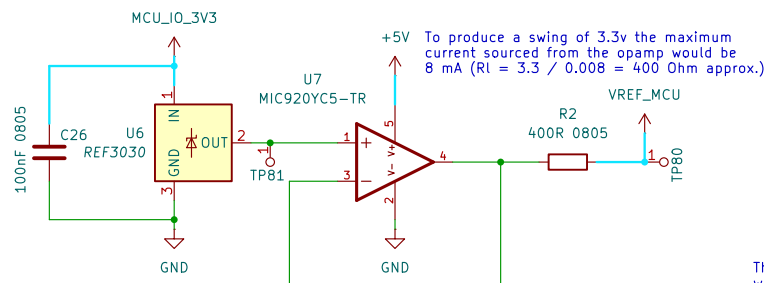
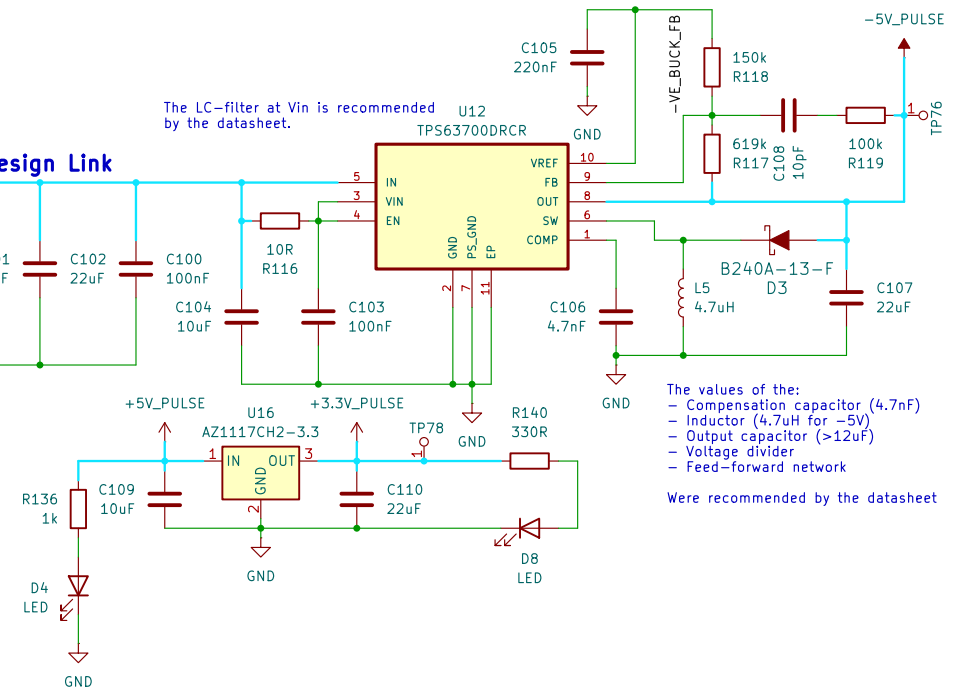
Output voltage from LNA+VGA:
 -1.125 -> 1.125 (2.25v swing)
 Input voltage range for the MCU's ADC:
 0 -> 3.3 (3.3v swing)
 The differential opamp configuration shifts the signal and amplifies it with a factor of 1.444 (3.9/2.7) to achieve a swing of 3.25.

VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.

In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.



The LC-filter at Vin is recommended by the datasheet.



These decoupling capacitor values were recommended by the datasheet